

Lecture 3: The Itô Integral

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3.1 Review of Riemann Calculus

This note presents the derivation of stochastic calculus beginning with simple Riemann summations. Classical Riemannian calculus for deterministic functions is generalized slightly for the construction of Riemann-Stieltjes integrals. We then construct an Itô stochastic integral, which has some similarities to the Riemann-Stieltjes integral, but fundamentally differs in the sense for which limit of the approximating summations converge to the integral.

In classical calculus, the Riemann integral $\int_0^T f(t)dt$ is the area under the curve of $f(t)$, and is constructed from approximations of the area under the curve using rectangles. There are backward and forward Riemann sums approximation,

$$\int_0^T f(s)ds \approx \sum_{i=0}^{n-1} f(t_i)(t_{i+1} - t_i) \quad (\text{backward})$$

$$\int_0^T f(s)ds \approx \sum_{i=0}^{n-1} f(t_{i+1})(t_{i+1} - t_i) \quad (\text{forward})$$

where the interval $[0, T]$ is partitioned by points $\{t_i\}_{i=0}^n$ with $0 = t_0 < t_1 < \dots < t_n = T$. Figure 3.1 illustrates.

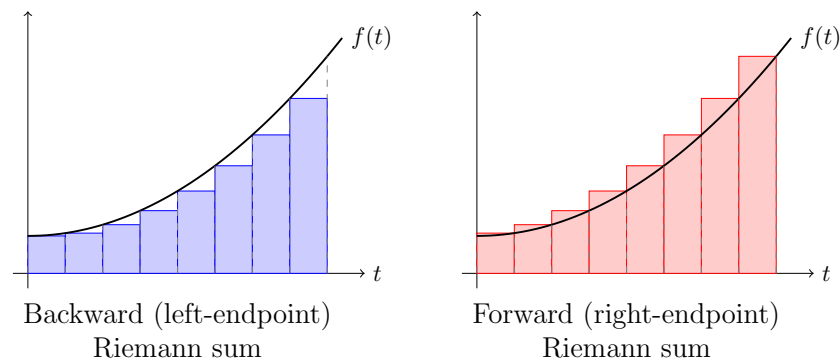


Figure 3.1: Forward and backward Riemann sum approximations of the area under curve $f(t)$.

We discussed quadratic variation in last lecture's presentation of Brownian motion. In this lecture we introduce total variation for the function f . Let Π denote a partition of $[0, T]$ with points $(t_i)_{i=0,1,\dots,n}$ with

$t_i \leq t_{i+1}$. The variation of f on this partition Π is

$$V(f, \Pi) = \sum_{i=0}^{n-1} |f(t_{i+1}) - f(t_i)| ,$$

and the total variation of f is the supremum over all possible partitions,

$$\text{TV}(f) = \sup_{\Pi} V(f, \Pi) .$$

If $\text{TV}(f) < \infty$ (i.e., f is function of **bounded variation**) then it can be represented as the difference of monotone-increasing functions,

$$f(t) = f^+(t) - f^-(t) ,$$

where $f^\pm(t + \delta) - f^\pm(t) \geq 0$ for any $\delta > 0$ at any point $t \in [0, T]$, and $\int_0^T f(t)dt = \int_0^T f^+(t)dt - \int_0^T f^-(t)dt$.

From Figure 3.1 we can see that a backward Riemann sum estimates $\int_0^T f^+(t)dt$ from below and forward Riemann sum from above, which gives us a squeeze lemma,

$$\sum_{i=0}^{n-1} f^+(t_i)(t_{i+1} - t_i) \leq \int_0^T f^+(t)dt \leq \sum_{i=0}^{n-1} f^+(t_{i+1})(t_{i+1} - t_i) .$$

For equally-spaced points $t_i = i\Delta t$ with $\Delta t = T/n$, the squeeze lemma can be simplified to give the Riemann sum's approximation error,

$$\sum_{i=0}^{n-1} f^+(t_{i+1})\Delta t - \sum_{i=0}^{n-1} f^+(t_i)\Delta t = \frac{T(f^+(T) - f^+(0))}{n} \rightarrow 0 \quad \text{as } n \rightarrow \infty .$$

In general, for any function $f(t)$ of bounded variation, a Riemann sum of the form $\sum_{i < n} f(t_i^*)(t_{i+1} - t_i)$ with $t_i^* \in [t_i, t_{i+1})$ and with $\sup_i(t_{i+1} - t_i) \rightarrow 0$ as $n \rightarrow \infty$, is an accurate and consistent approximation of $\int_0^T f(t)dt$.

3.1.1 Differentiation

For any $t \in [0, T]$ define the function

$$F(t) = \int_0^t f(s)ds ,$$

and by Fundamental Theorem of Calculus,

$$F'(t) = f(t) .$$

For the small step-size $\Delta t_i = t_{i+1} - t_i$, we can approximate the change in F with a Taylor expansion,

$$F(t_{i+1}) - F(t_i) = f(t_i)\Delta t_i + \frac{1}{2}f'(t_i)\Delta t_i^2 + R(t_i, t_{i+1}) ,$$

where $R(t_i, t_{i+1})$ is a remainder time of higher-order (i.e., $R(t_i, t_{i+1}) = \mathcal{O}(\Delta t_i^3)$). This Taylor expansion is useful for approximating the change in $F(t)$, but under the a summation, as $\max_{i < n} \Delta t_i$ shrinks toward zero, it is only the 1st-order terms (i.e., terms of order Δt_i) that matter,

$$\begin{aligned} \left| F(T) - F(0) - \sum_{i=0}^{n-1} f(t_i)\Delta t_i \right| &= \left| \sum_{i=0}^{n-1} (F(t_{i+1}) - F(t_i)) - \sum_{i=0}^{n-1} f(t_i)\Delta t_i \right| \\ &= \left| \sum_{i=0}^{n-1} \left(\frac{1}{2}f'(t_i)\Delta t_i^2 + R(t_i, t_{i+1}) \right) \right| \\ &\leq \frac{\max_{i < n} \Delta t_i}{2} \sum_{i=0}^{n-1} |f'(t_i)| \Delta t_i + \mathcal{O}(T(\max_{i < n} \Delta t_i)^2) \\ &\rightarrow 0 , \end{aligned}$$

as $n \rightarrow \infty$. This seems almost by construction that the integrand is the function's derivative, but as we proceed to construct stochastic calculus we will see a 2nd-order term remaining after we take the limit, thus requiring us to adjust how we take a derivative in the stochastic setting.

3.1.2 Equivalence of Integrals and Differential Equations

In Riemann calculus we can write an integral equation,

$$X_t = X_0 + \int_0^t a(X_s) ds ,$$

or equivalently we can write in differentiated form,

$$\frac{d}{dt} X_t = a(X_t) ,$$

with initial condition X_0 . There are a variety of other ways to denote differentiated form, all of which are equivalent,

$$\begin{aligned} \dot{X}_t &= a(X_t) \\ X'_t &= a(X_t) \\ dX_t &= a(X_t) dt . \end{aligned}$$

3.1.3 Riemann-Stieltjes Integrals

Consider some functions $g(t)$ and $F(t)$. If $F(t)$ has bounded variation on $[0, T]$, then we can define the Riemann-Stieltjes integral as

$$I[g] := \int_0^T g(t) dF(t) = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} g(t_i^*) (F(t_{i+1}) - F(t_i)) , \quad (3.1)$$

where the points $(t_i)_{i=0,1,\dots,n}$ form a partition of $[0, T]$ with $\max_{i < n} (t_{i+1} - t_i) \rightarrow 0$ as $n \rightarrow \infty$, and with $t_i^* \in [t_i, t_{i+1})$. In (3.1), $g(t)$ is the integrand and $F(t)$ is the integrating (signed) measure. The Stieltjes integral allows for non-differentiability of F . If there is a derivative $F'(t) = f(t)$ then the Stieltjes integral is equivalent to a Riemann integral, then $I[g] = \int_0^T g(t) f(t) dt$. However, sometimes $F(t)$ could be non-differentiable in the traditional Riemann sense, that is,

$$\lim_{h \rightarrow 0} \frac{F(t+h) - F(t)}{h} ,$$

may not exist for some open interval of the real line, but even without this limit the Stieltjes integral can be still well defined so long as F has finite total variation.

Taking $F(t) = t$ in (3.1) is equivalent to the standard Riemann integral from earlier.

The Riemann-Stieltjes integral has a natural usage in probability theory because $F(t)$ could be a cumulative distribution function (CDF) for a random variable in $[0, T]$. For $F(t)$ a CDF for random variable $\tau \in [0, T]$, then $\mathbb{E}g(\tau) = \int_0^T g(t) dF(t)$. Moreover, if $F'(t) = f(t)$, then $f(t)$ is a probability density and $\mathbb{E}g(\tau) = \int_0^T g(t) f(t) dt$.

The Riemann-Stieltjes integral can also be constructed using a path X_t ,

$$I[g] = \int_0^T g(X_s) dF(X_s) = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} g(X_{t_i}) (F(X_{t_{i+1}}) - F(X_{t_i})) . \quad (3.2)$$

If there are derivatives $dX_t = a(X_t) dt$ and $F'(x) = f(x)$, then $I[g] = \int_0^T g(X_t) f(X_t) a(X_t) dt$.

3.2 Stochastic Integrals

On a probability space $(\Omega, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, let W_t be a standard Brownian motion, and consider a sequence of partitions $\Pi^n = \{t_i : i = 0, 1, \dots, n\}$ of $[0, t]$ with $|\Pi^n| \rightarrow 0$ as $n \rightarrow \infty$. Motivated by the Stieltjes integral in (3.2) with $X_t = W_t$ and $F(x) = x$, for g bounded function we can define an **Itô stochastic integral**:

$$I_t[g] = \int_0^t g(W_s) dW_s = \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n} g(W_{t_i})(W_{t_{i+1}} - W_{t_i}), \quad (3.3)$$

where here **it is essential that the integral is evaluated at the back of the interval**. The integral in (3.3) is fundamentally different from the Riemann-Stieltjes integral because W_t does not have bounded variation; the Riemann-Stieltjes does not exist if the integrator does not have bounded variation. However, properties of the Brownian motion increments all us to prove a limit for the summation in (3.3), in the mean-square sense. In particular, we show the summations on the right-hand side of (3.3) are a Cauchy sequence in mean square. For partition sizes n and m , consider the union of partitions $\Pi^n \cup \Pi^m$, and for $t_i \in \Pi^n \cup \Pi^m$ let $t_i^{(n)} = \sup\{t \in \Pi^n : t \leq t_i\}$ and $t_i^{(m)} = \sup\{t \in \Pi^m : t \leq t_i\}$; Figure 3.2 illustrates how points $t_i^{(n)}$ and $t_i^{(m)}$ ensure the integrand is evaluated without anticipating the Brownian increment on $[t_i, t_{i+1})$. If g has countably-many discontinuities then

$$\begin{aligned} & \mathbb{E} \left(\sum_{t_i \in \Pi^n} g(W_{t_i})(W_{t_{i+1}} - W_{t_i}) - \sum_{t_i \in \Pi^m} g(W_{t_i})(W_{t_{i+1}} - W_{t_i}) \right)^2 \\ &= \mathbb{E} \left(\sum_{t_i \in \Pi^n \cup \Pi^m} \left(g(W_{t_i^{(n)}}) - g(W_{t_i^{(m)}}) \right) (W_{t_{i+1}} - W_{t_i}) \right)^2 \\ &= \sum_{t_i \in \Pi^n \cup \Pi^m} \mathbb{E} \left(g(W_{t_i^{(n)}}) - g(W_{t_i^{(m)}}) \right)^2 (t_{i+1} - t_i) \\ &\rightarrow 0, \end{aligned}$$

as $n, m \rightarrow \infty$. This proves the existence of a mean-square limit of the summations on the right-hand side of (3.3), and hence the Itô integral is defined to be the random variable that is this limit.

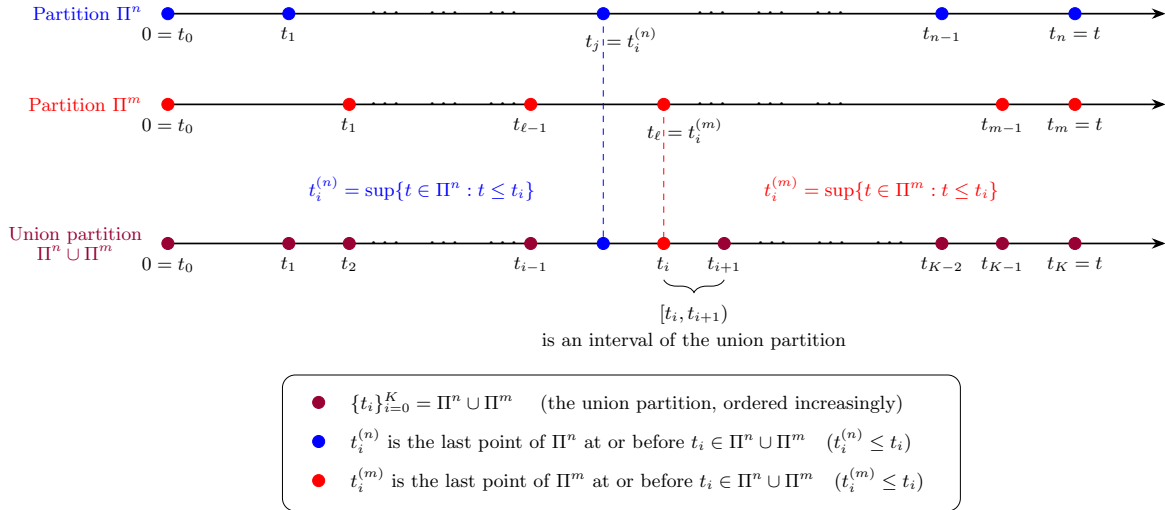


Figure 3.2: The points $t_i^{(n)}$ and $t_i^{(m)}$ are selected to ensure the integrand is evaluated without anticipating the Brownian increment on $[t_i, t_{i+1})$.

3.2.1 General Itô Integrals

For unbounded g such that $\mathbb{E} \int_0^t |g(W_s)|^2 ds < \infty$ (i.e., $g(W_t)$ is square integrable), the construction of the Itô integral is similar to (3.3), but there are some technical issues that need to be handled. Moreover, the Itô integral in (3.3) can be generalized for any integrand g_t that is adapted to $\mathcal{F}_t$ with $\mathbb{E} \int_0^t g_s^2 ds < \infty$.

3.2.2 Basic Properties for General Itô Integrals

For any pair of functions g_t and f_t adapted to $\mathcal{F}_t$ with $\mathbb{E} \int_0^t g_s^2 ds < \infty$ and $\mathbb{E} \int_0^t f_s^2 ds < \infty$, there are the following are properties of the Itô integral:

- (Linearity) For any scalars λ and α ,

$$I_t[\lambda g + \alpha f] = \lambda I_t[g] + \alpha I_t[f] .$$

- (Additivity) For $0 \leq s \leq t$,

$$I_t[g] = I_s[g] + \int_s^t g_u dW_u .$$

- (Mean Zero)

$$\mathbb{E} I_t[g] = 0 .$$

- (Itô Isometry)

$$\mathbb{E} [I_t[g] I_t[f]] = \int_0^t \mathbb{E} [g_s f_s] ds .$$

- (Martingale Property) For $0 < s < t \leq T$,

$$\mathbb{E} [I_t[g] | \mathcal{F}_s] = I_s[g] .$$

These five basic properties can be deduced from the statistics of a finite summation like that on the right-hand side of (3.3).

Finally, if the integrand is deterministic then Itô integral is normally distributed. That is, for $g(t)$ non-stochastic and only a function of time,

$$\int_0^t g(s) dW_s \sim \text{normal} \left(0, \int_0^t g^2(s) ds \right) . \quad (3.4)$$

In general, Itô integrals are non-normal, for example $\int_0^t W_s dW_s$ which we will see in the next lecture.

3.2.3 Stratonovich-Type Integrals

At the beginning of this note we began constructing Riemann sums using the backward point in the interval (i.e., $\int f(t) dt \approx \sum_i f(t_i)(t_{i+1} - t_i)$), which are more precisely referred to as backward Riemann sums. There are of course forward Riemann sums (i.e., $\sum_i f(t_{i+1})(t_{i+1} - t_i)$), and there are more general sums where the integrand is evaluated at any point in the interval (i.e., $\sum_i f(t_i^*)(t_{i+1} - t_i)$ where $t_i^* \in [t_i, t_{i+1}]$). For classical Riemann-Stieltjes it does not matter where in the interval that the integrand is evaluated. However, for stochastic integrals it matters because taking a forward sum could cause the integrand to be anticipating, in which case we no longer have the mean-zero property or the martingale property. The

stochastic integral arising from a non-backward-point summation are the midpoint $t_i^* = (t_{i+1} + t_i)/2$ is referred to as a Stratonovich integral, and is denoted with a $\circ$,

$$\int_0^t g_t \circ dW_t = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} g_{t_i^*} (W_{t_{i+1}} - W_{t_i}) .$$

Stratonovich integrals are of limited use to us in finance, but occasionally they come up (for example, in Monte Carlo simulation of the CIR model for short interest rates).

Exercise: show for $g_t = g(W_t)$ that it is equivalent to define the Stratonovich as an averaged integrand, $\int_0^t g(W_t) \circ dW_t = \lim_{n \rightarrow \infty} \sum_i \frac{1}{2} (g(W_{t_{i+1}}) + g(W_{t_i})) (W_{t_{i+1}} - W_{t_i})$.

3.3 Remarks

1. A good background reference for this material is [1] and [2].
2. For a more rigorous treatment, see [3].

References

- [1] T. Björk. *Arbitrage Theory in Continuous Time*. Oxford University Press, Oxford, UK, 4th edition, 2020.
- [2] B. Oksendal. *Stochastic differential equations: an introduction with applications*. Springer Science & Business Media, 2013.
- [3] S. E. Shreve and I. Karatzas. Brownian motion and stochastic calculus. *Graduate texts in mathematics*, 113, 1991.